

That's big!

ZSR

$$y(t) = x(t) * h(t)$$

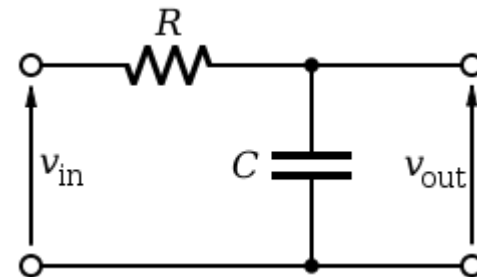
SYDE252 - W05A - Zero State Response
Dr. Matt Borland

For linear systems, the response of the system is the sum of ZIR and ZSR:

$$\text{Total Response} = \text{Zero-Input Response} + \text{Zero-State Response}$$

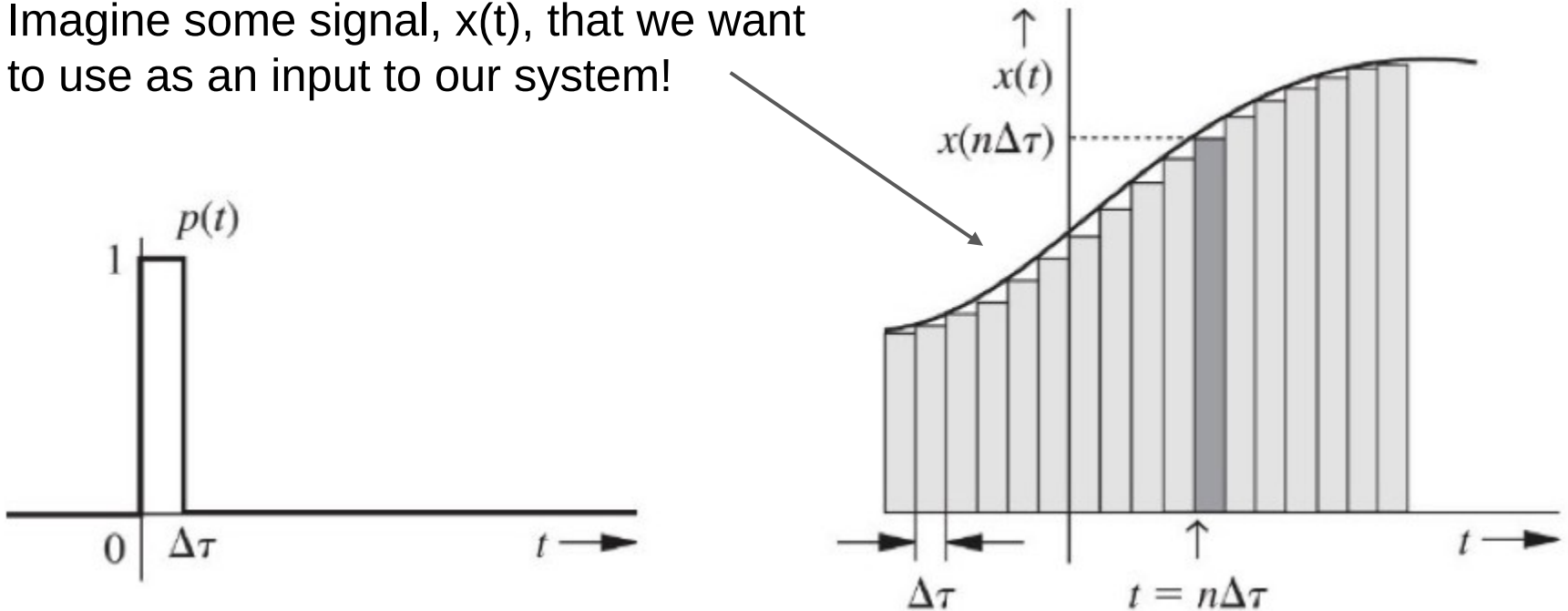
The zero-input response is the system output when $x(t) = 0$.

The zero-state response is the system output to the external input $x(t)$ when the system is in zero state, meaning the absence of all internal energy storages: that is, all initial conditions are zero.

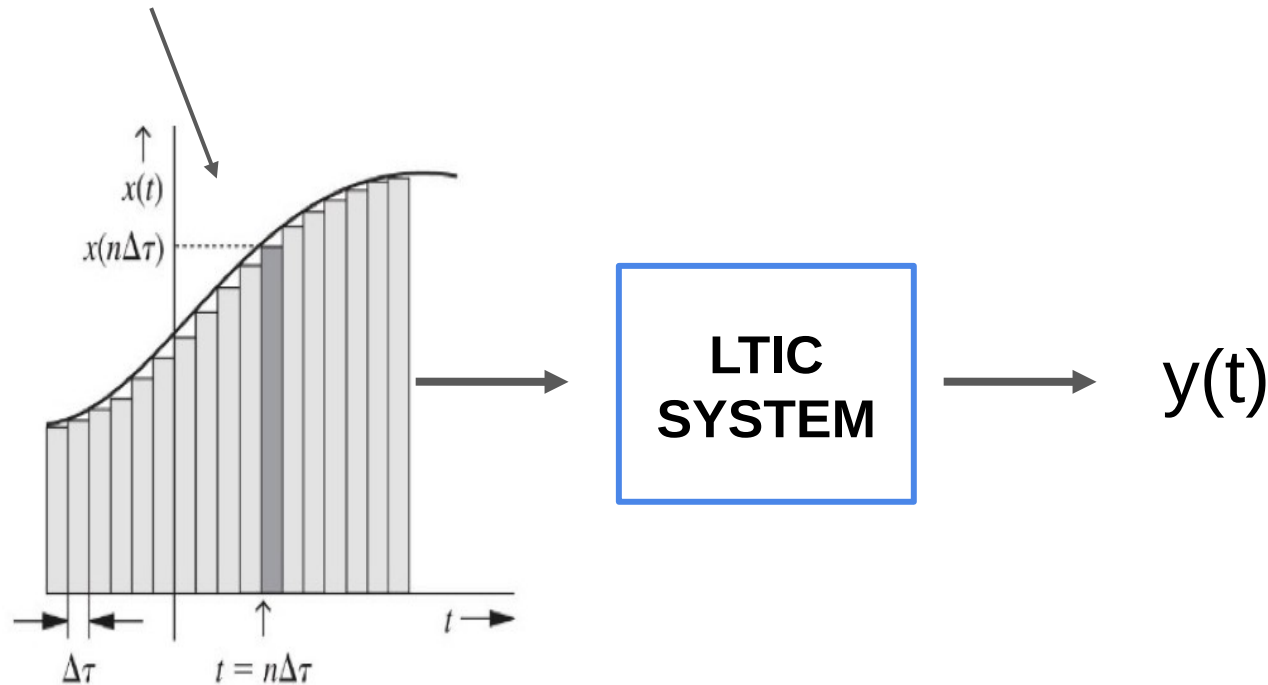


The ZSR is the system response to an external input when the IC's are zero:

Imagine some signal, $x(t)$, that we want to use as an input to our system!

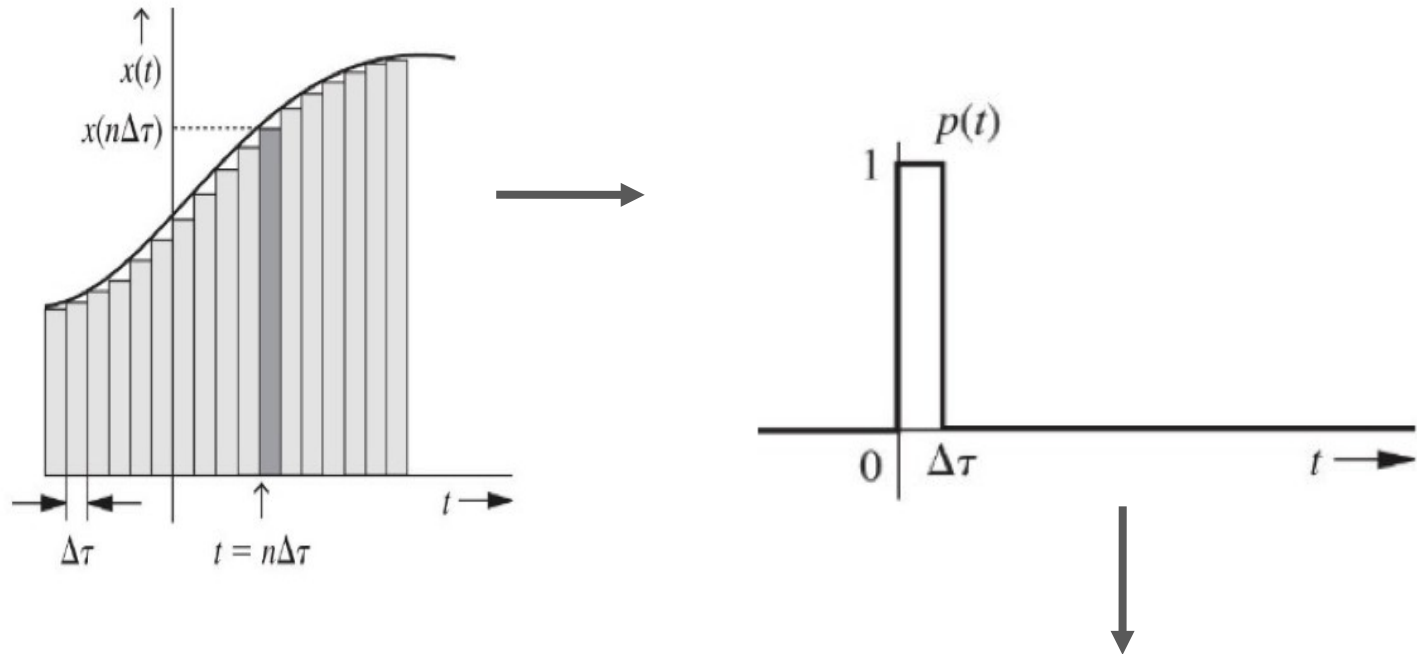


Imagine some signal, $x(t)$, that we want to use as an input to our system!

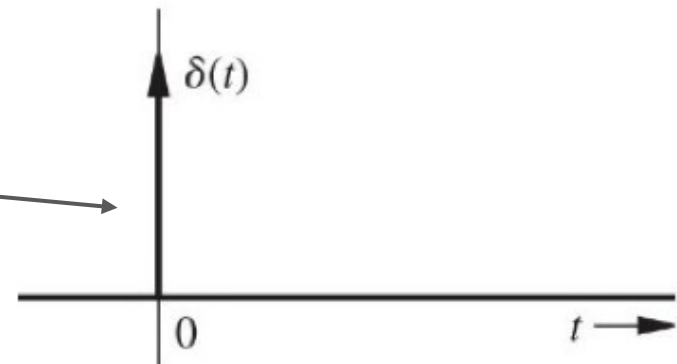


...but we know it's an LTIC system, so we can just break $x(t)$ into a bunch of simpler chunks ($x_1(t)$, $x_2(t)$, etc.), push those through the system to get ($y_1(t)$, $y_2(t)$, etc.), and then add all of those up to get our overall $y(t)$.

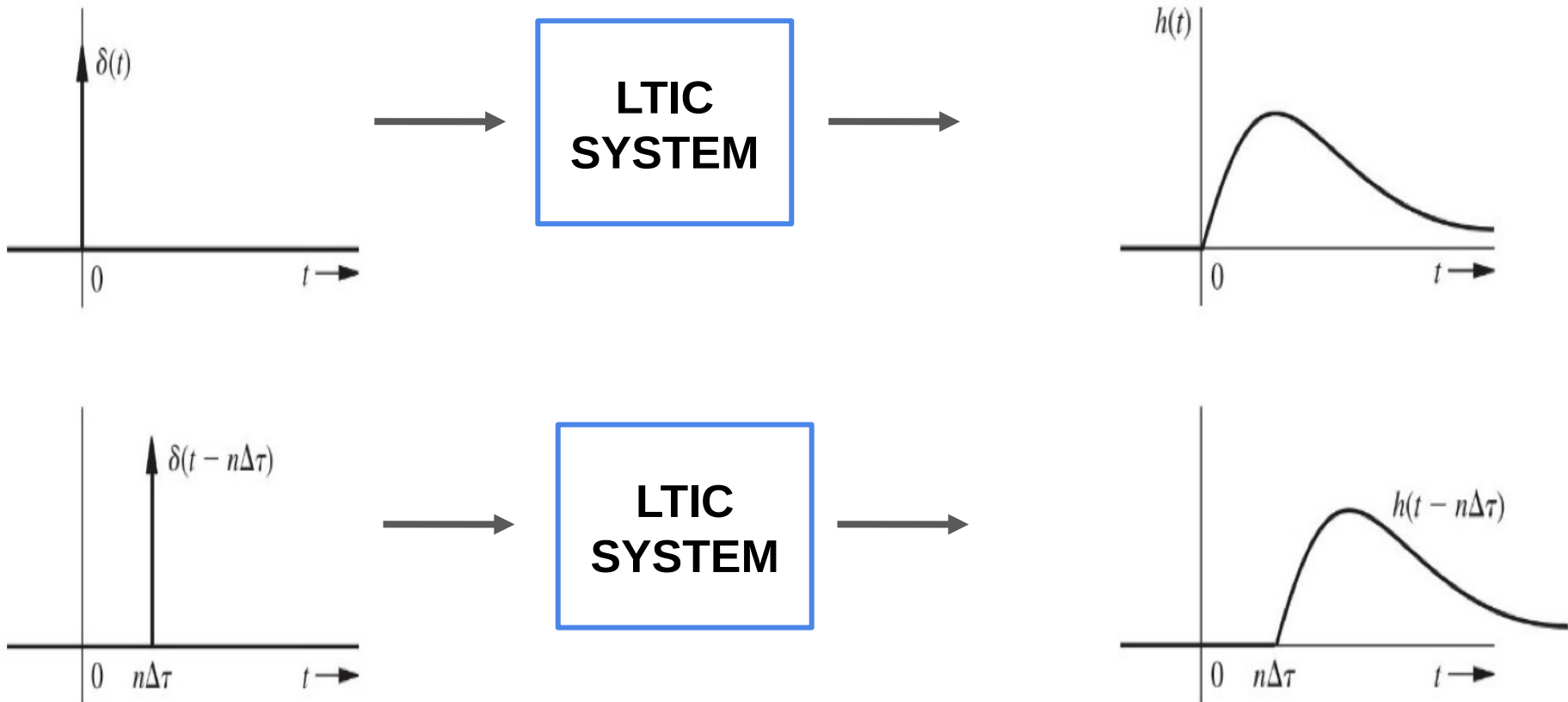
Let's turn our $x(t)$ into a bunch of scaled and time shifted $p(t)$'s!



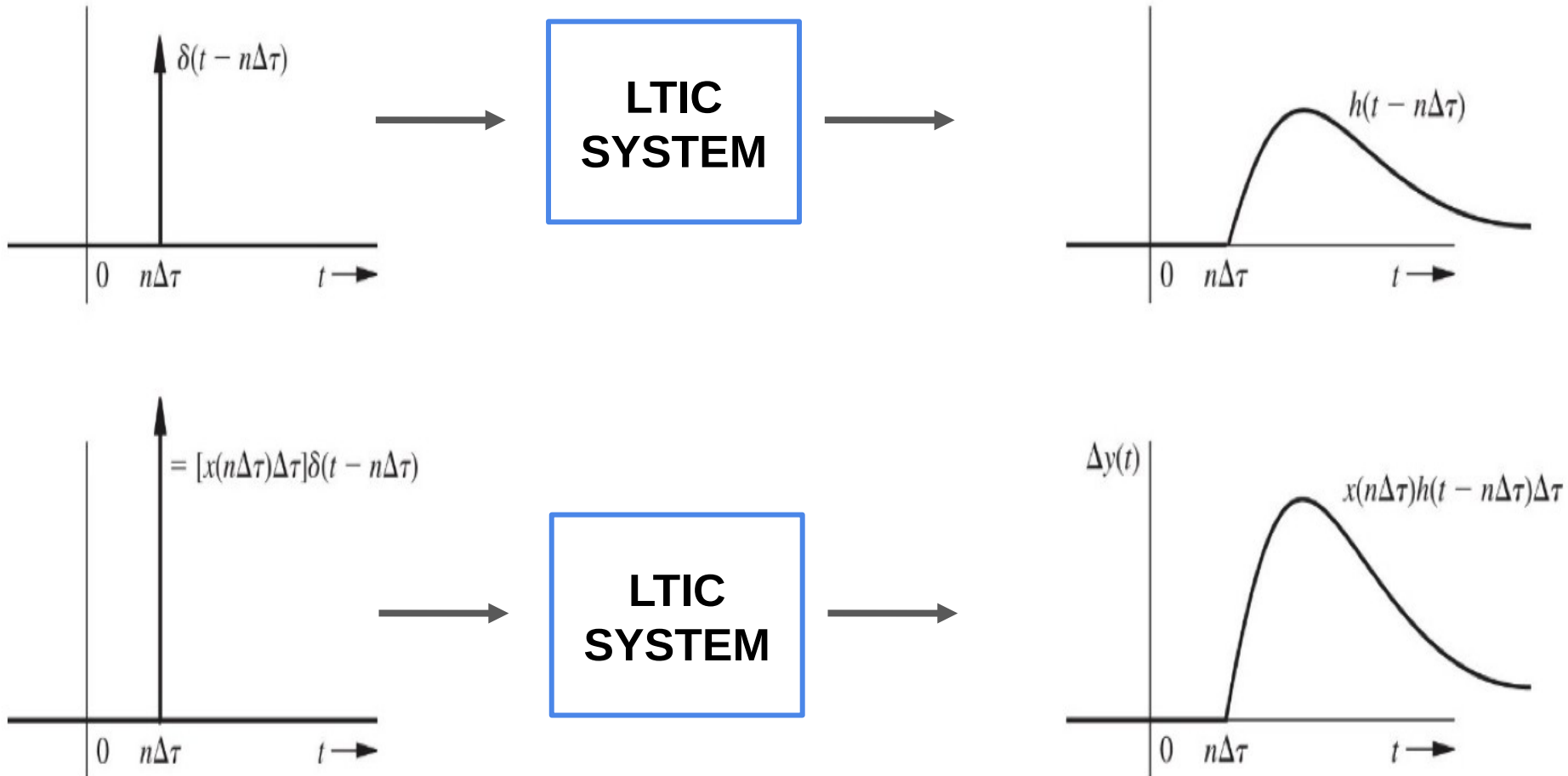
We can also think about making those $p(t)$'s really skinny to become similar to

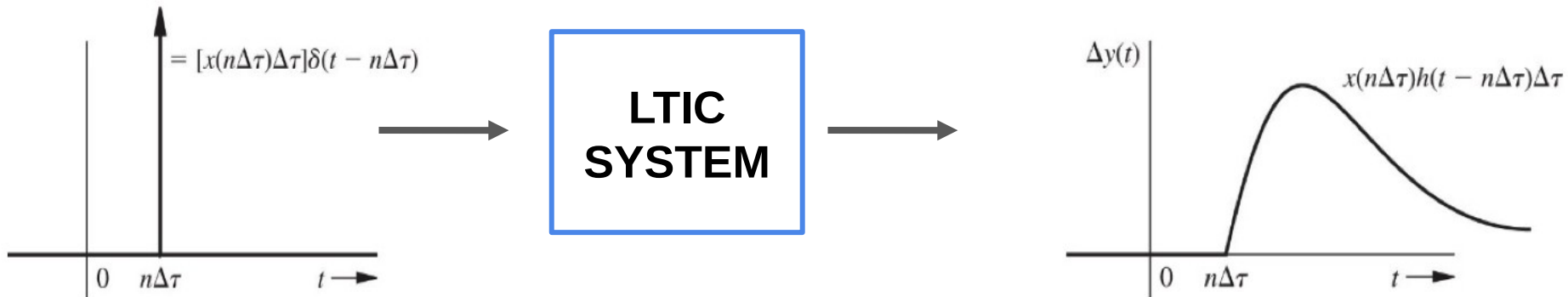


From our derivation of $h(t)$, the unit impulse response, we know that for a LTIC system:

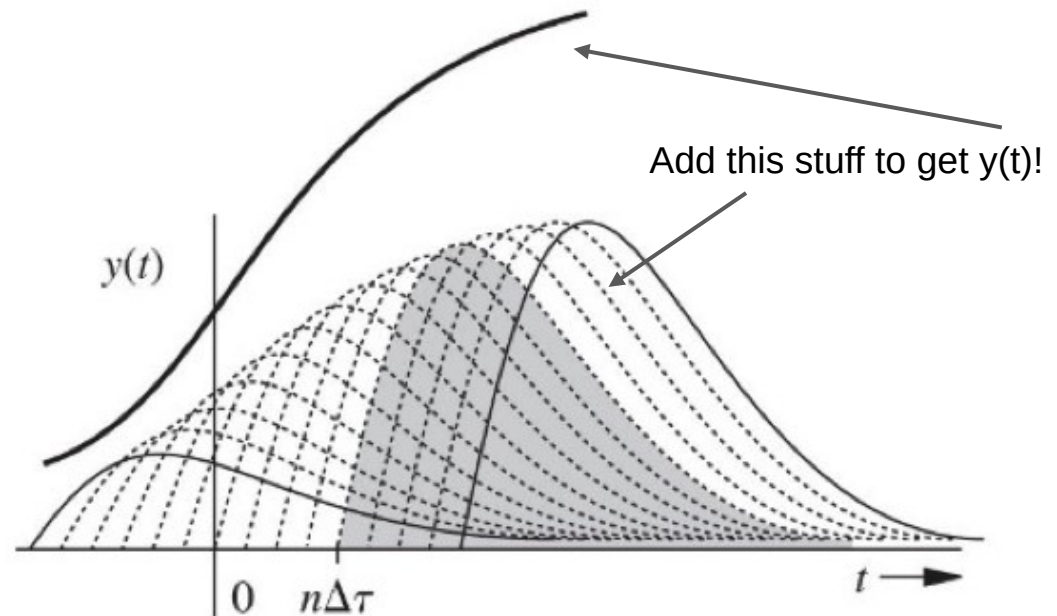


So now let's take the smallest possible slice of $x(t)$ at time $n\Delta\tau$

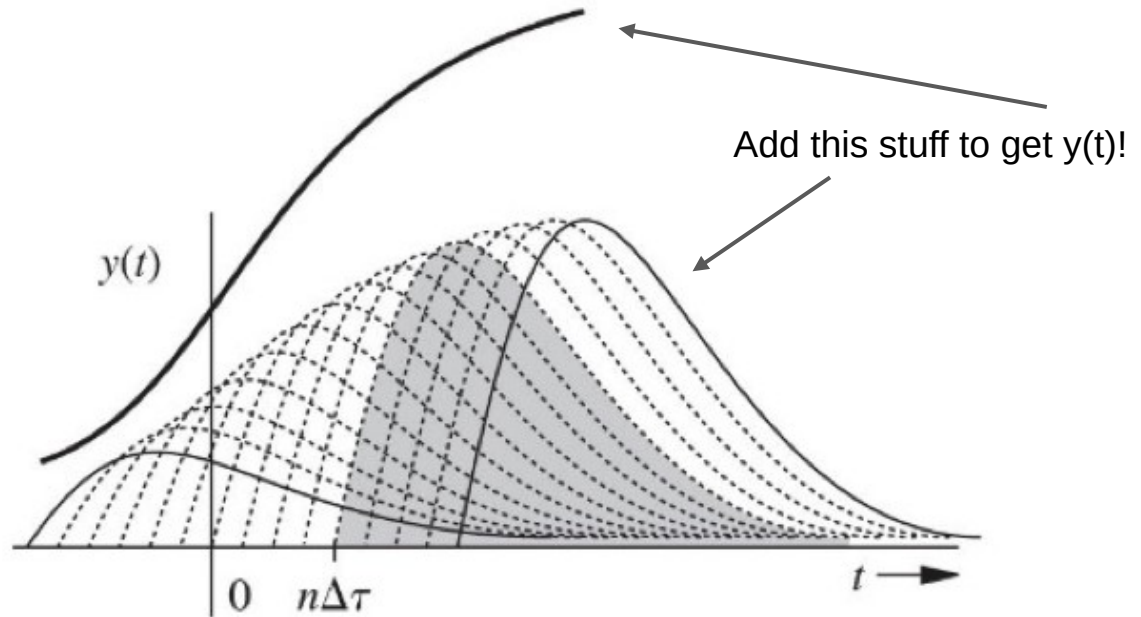




So let's do this for all possible $n\Delta\tau$'s and then add all of those them up to get $y(t)$:



More formally:



$$y(t) = \lim_{\Delta\tau \rightarrow 0} \sum_{\tau} x(n\Delta\tau) h(t - n\Delta\tau) \Delta\tau$$

Which is:

$$= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

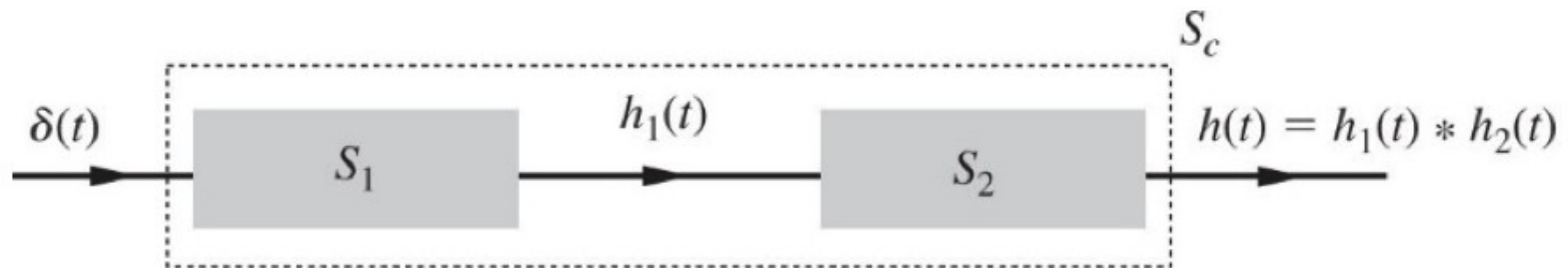
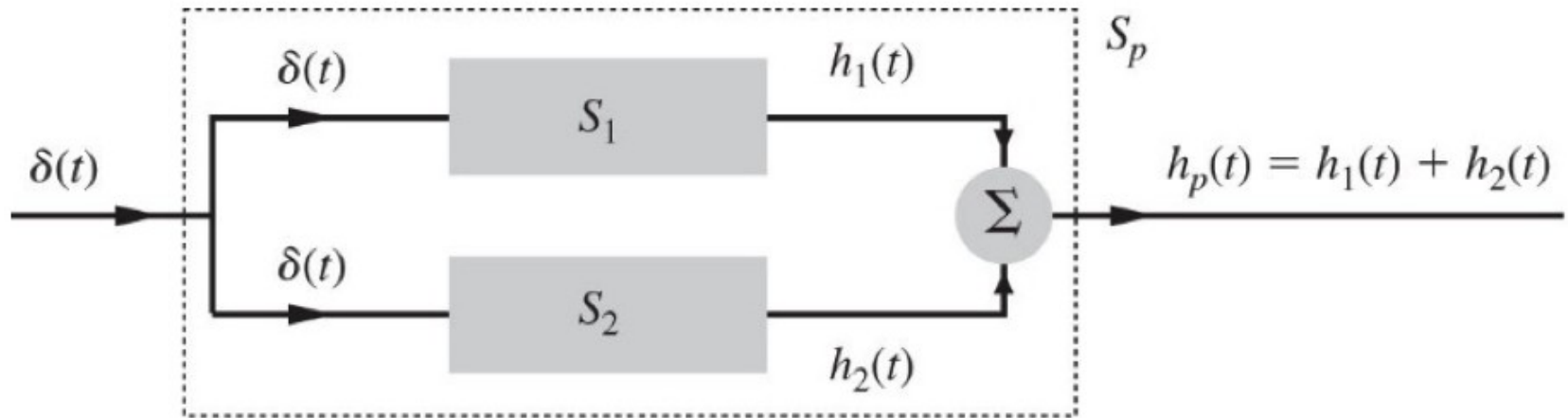
More formally:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

This is how we defined convolution earlier!

$$y(t) = x(t) * h(t)$$

Interconnected systems! Remember our convolution rules :)



A special function for LTIC systems - let's get complex!



Where 's' is a complex variable...this means we have an **everlasting** sinusoid!

$$y(t) = h(t) * e^{st} = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

If we define:

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

then:

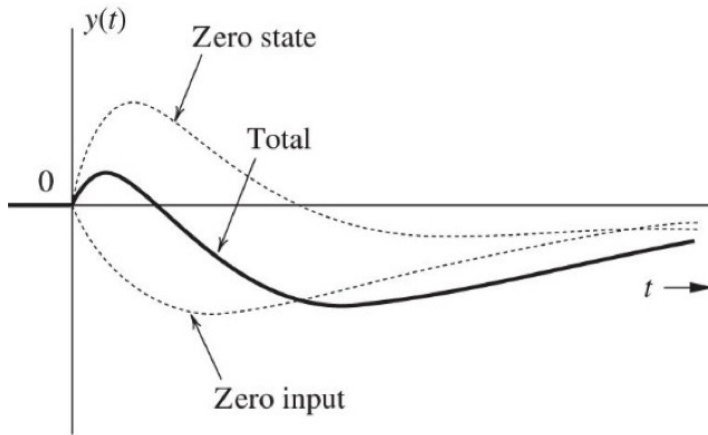
$$y(t) = H(s) e^{st}$$

$$H(s) = \frac{y(t)}{e^{st}} \left. \begin{array}{l} \text{OUTPUT} \\ \text{INPUT} \end{array} \right\} \text{TRANSFER FUNCTION}$$

Laplace is coming, Sansa Stark.

Total System Response!

$$\text{total response} = \underbrace{\sum_{k=1}^N c_k e^{\lambda_k t}}_{\text{ZIR}} + \underbrace{x(t) * h(t)}_{\text{ZSR}}$$



ZIR

The response of the system with no input applied, but (potentially) with energy stored in it at time zero.

ZSR

The response of the system to an applied input signal when the system is initially at rest (i.e. IC's = 0).

NOTE: ZIR is a slightly different form when you have repeated roots.